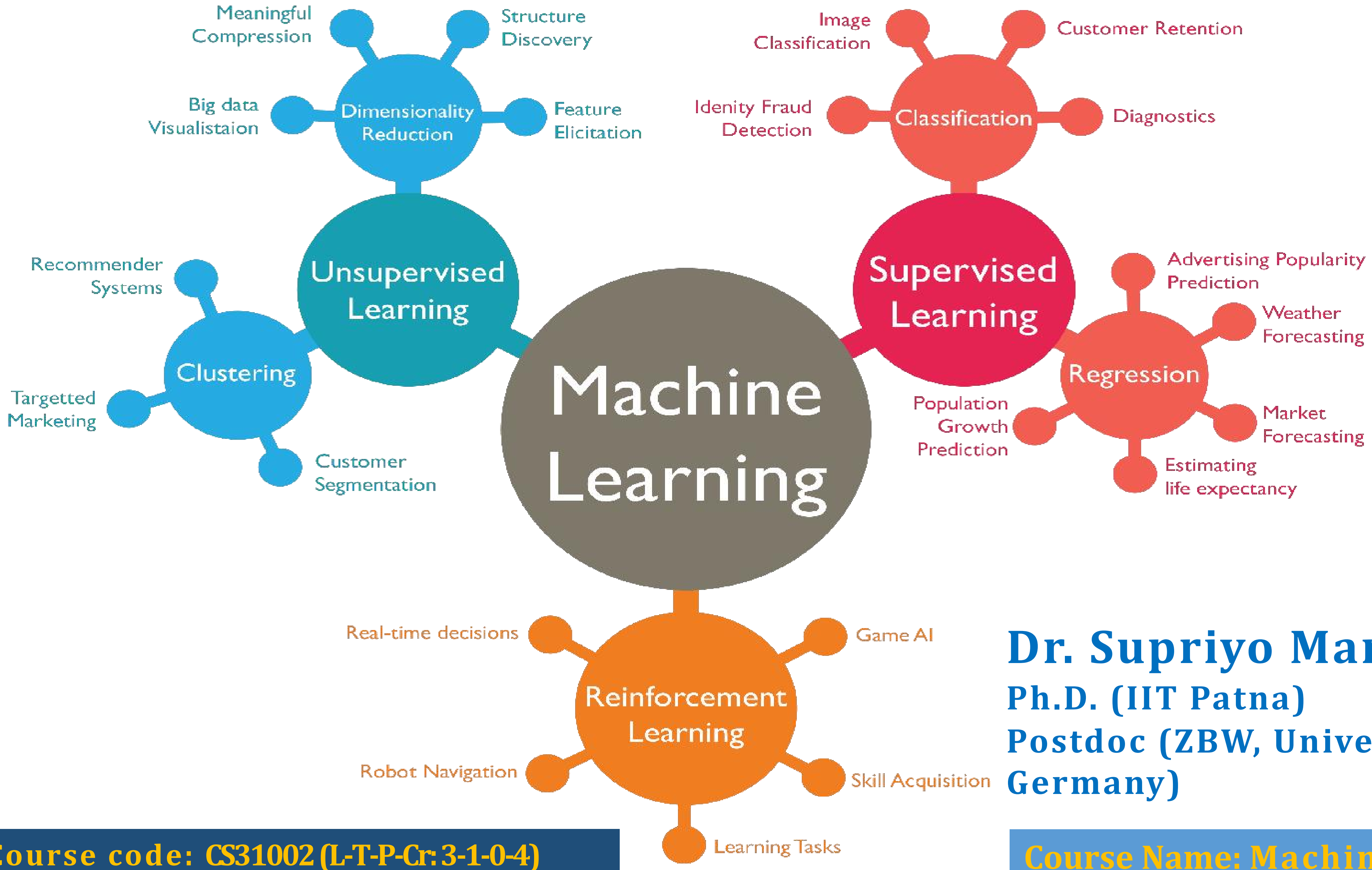

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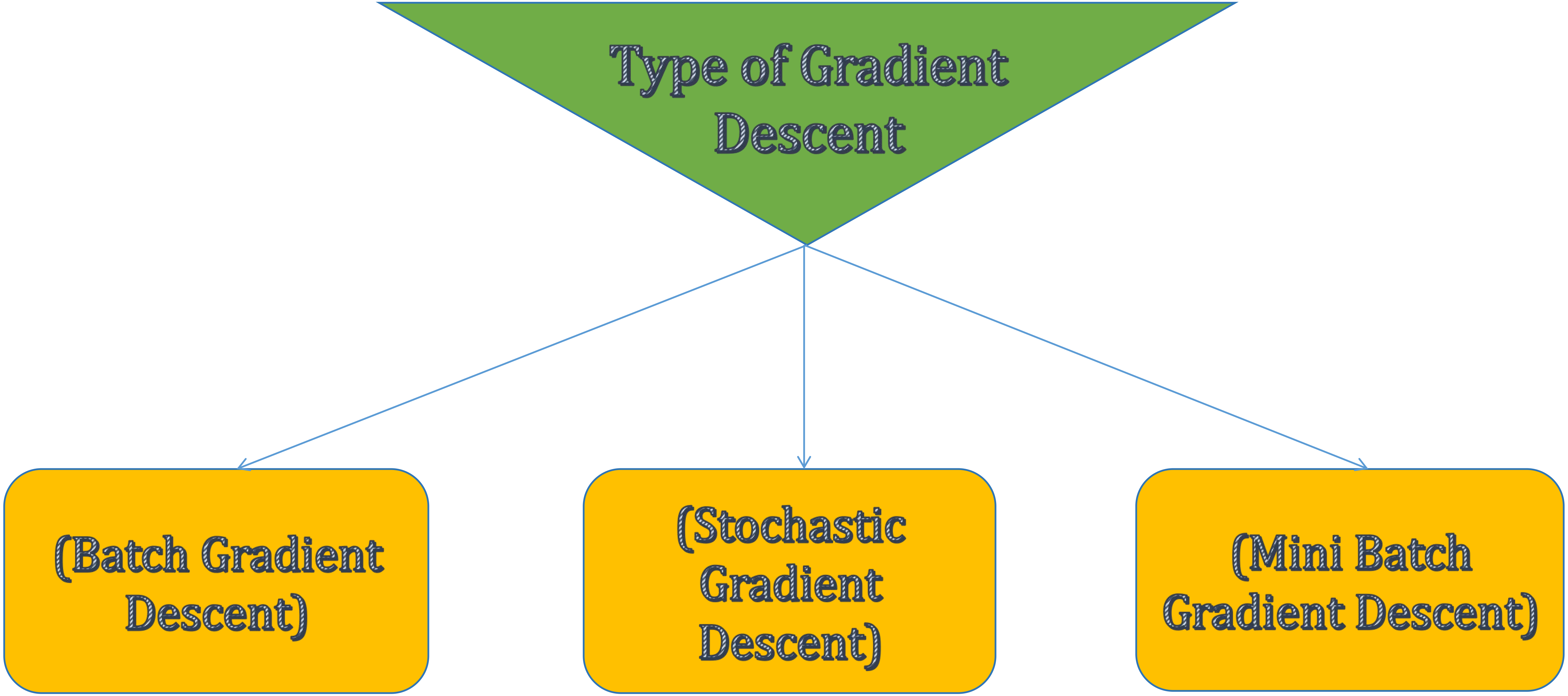
Credits: 4

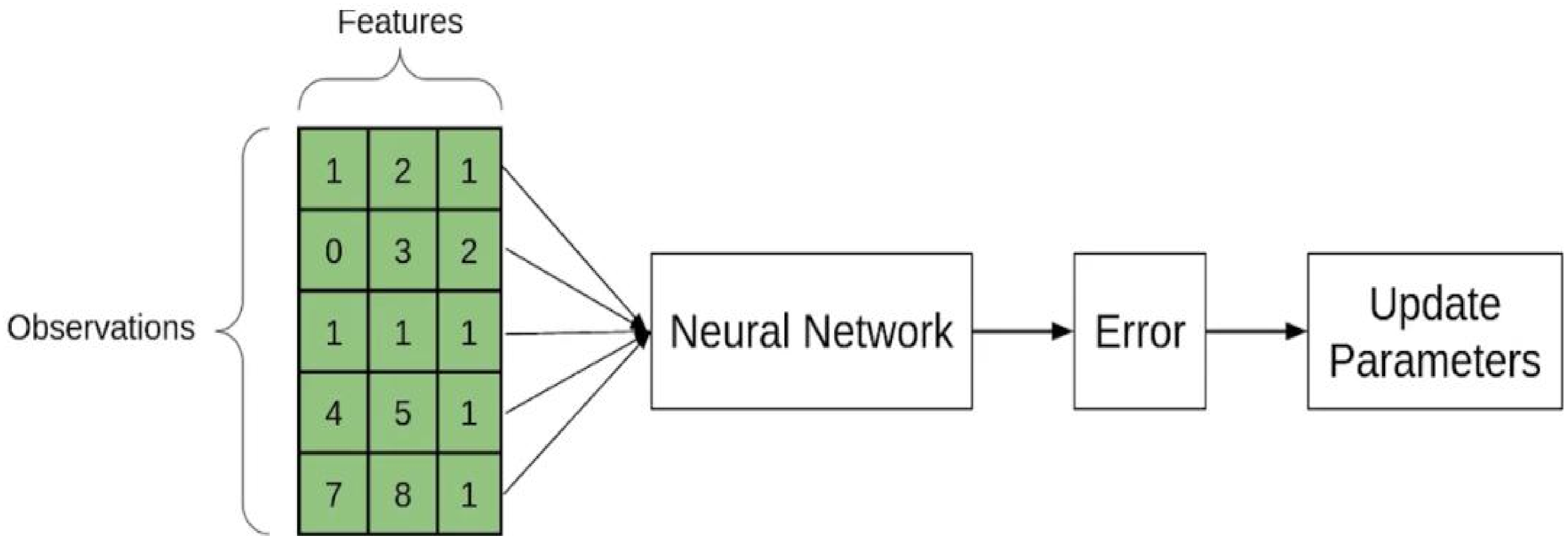
Course Name: Machine Learning


Type of Gradient Descent
(Batch Gradient Descent)
(Stochastic Gradient Descent)
(Mini Batch Gradient Descent)

Batch Gradient Descent

- In Batch Gradient Descent, all the training data is taken into consideration to take a single step.
- We take the average of the gradients of all the training examples and then use that mean gradient to update our parameters. So that's just one step of gradient descent in one epoch.
- In batch Gradient Descent since we are using the entire training set, the parameters will be updated only once per epoch.
- Batch Gradient Descent is great for convex or relatively smooth error manifolds.
- In this case, we move somewhat directly towards an optimum solution.
- The graph of cost vs epochs is also quite smooth because we are averaging over all the gradients of training data for a single step. The cost keeps on decreasing over the epochs.

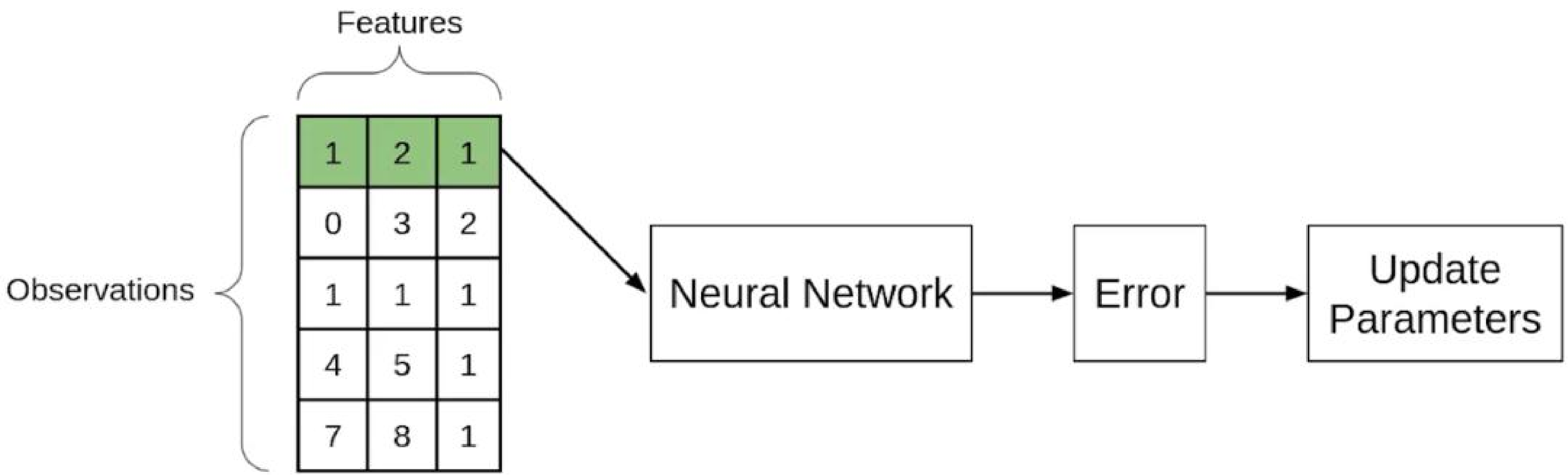
Batch Gradient Descent



Stochastic Gradient Descent

- ❖In Batch Gradient Descent we were considering all the examples for every step of Gradient Descent. But what if our dataset is very huge.
- ❖Suppose our dataset has 5 million examples, then just to take one step the model will have to calculate the gradients of all the 5 million examples.
- ❖This does not seem an efficient way. To tackle this problem we have Stochastic Gradient Descent.
- ❖In Stochastic Gradient Descent (SGD), we consider just one example at a time to take a single step.
- ❖Since we are considering just one example at a time the cost will fluctuate over the training examples and it will not necessarily decrease. But in the long run, you will see the cost decreasing with fluctuations.
- ❖Because the cost is so fluctuating, it will never reach the minima but it will keep dancing around it.
- ❖SGD can be used for larger datasets. It converges faster when the dataset is large as it causes updates to the parameters more frequently.

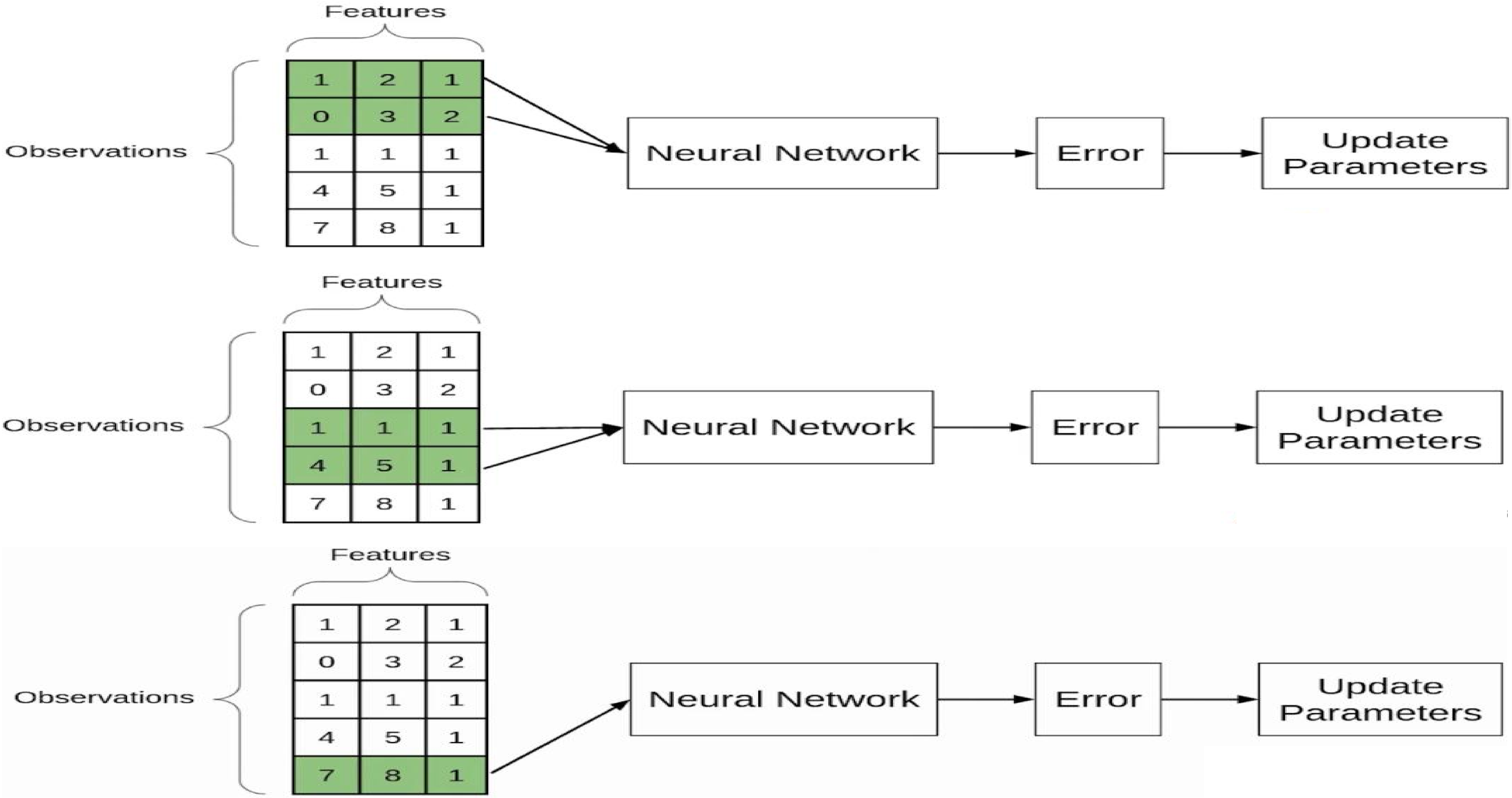
Stochastic Gradient Descent



Mini Batch Gradient Descent

- Batch Gradient Descent converges directly to minima. SGD converges faster for larger datasets. But, since in SGD we use only one example at a time, we cannot implement the vectorized implementation on it.
- This can slow down the computations. To tackle this problem, a mixture of Batch Gradient Descent and SGD is used.
- Neither we use all the dataset all at once nor we use the single example at a time.
- We use a batch of a fixed number of training examples which is less than the actual dataset and call it a mini-batch. Doing this helps us achieve the advantages of both the former variants (GD and SGD).
- Just like SGD, the average cost over the epochs in mini-batch gradient descent fluctuates because we are averaging a small number of examples at a time.
- When we are using the mini-batch gradient descent we are updating our parameters frequently as well as we can use vectorized implementation for faster computations.

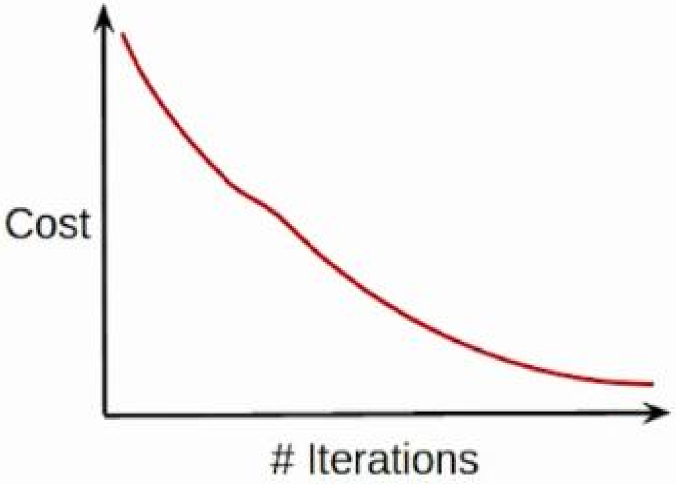
Mini Batch Gradient Descent



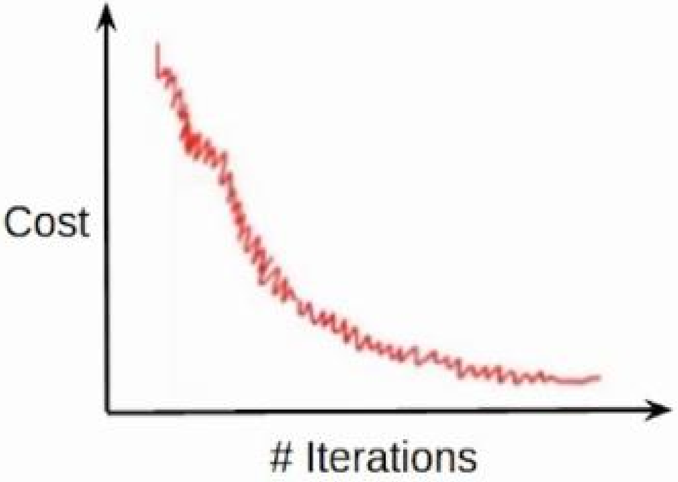
Comparison between different type of Gradient Descent

Batch Gradient Descent	Stochastic Gradient Descent (SGD)	Mini-Batch Gradient Descent
• Entire dataset for updation	• Single observation for updation	• Subset of data for updation
• Cost function reduces smoothly	• Lot of variations in cost function	• Smoother cost function as compared to SGD
• Computation cost is very high	• Computation time is more	• Computation time is lesser than SGD
		• Computation cost is lesser than Batch Gradient Descent

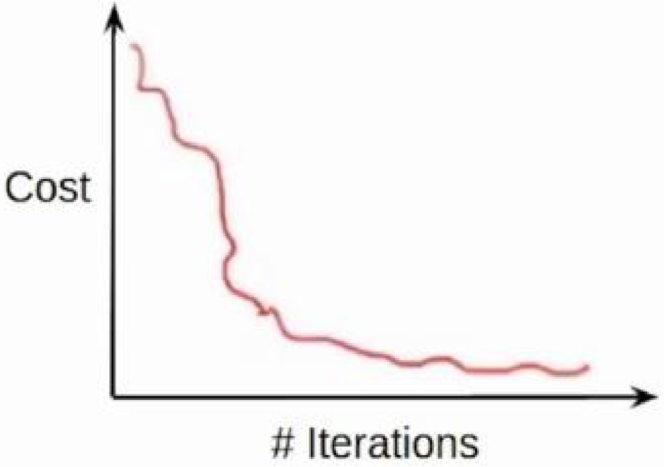
- Cost function reduces smoothly



- Lot of variations in cost function



- Smoother cost function as compared to SGD



Problem..........

Negative feedback	Positive feedback	Popularity score
1	1	3.25
1	2	6.5
2	2	3.5
0	1	5.0

Apply Batch Gradient Descent, Stochastic Gradient Descent and Mini-batch Gradient Descent (mini batch =2) on loss function of linear regression method using Mean Square Error.

Suppose, intercept=1, slope(s) are -1 and 2 respectively, learning rate=$\mathbf{0.0001}$.
After first epoch, what will be value of intercept, slopes of different gradient descent technique?